

Equivariant magnitude, Borel constructions, and distance algebras

Research note — 15 September 2026

Abstract

We compare the representation-valued equivariant refinement of magnitude (co)homology, its Borel version, and their realizations over skew group algebras of distance algebras. We then construct an integral Burnside-ring-valued magnitude series and describe a possible refinement using Bredon (co)homology. Ground-ring assumptions, module structures, and the difference between fixed points and invariant vectors are kept explicit. Ordinary magnitude (co)homology and the Asao–Ivanov algebraic model are assumed throughout. Results taken from the literature are identified at their point of use; equivariant specializations and additional deductions are proved here. No claim of novelty is intended.

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1 The usual representation-valued equivariant theory

1.1 Standing conventions and coefficients

Let X be a finite metric space, and let G be a finite group acting on X by isometries. Graphs are treated through their vertex metrics; the letter G always denotes the symmetry group. Unless explicitly stated otherwise, all distances are finite. The empty metric space is allowed as a fixed-point space, and its magnitude and all its magnitude groups are zero. Every fixed subset carries the *restricted metric*. In particular, for a graph, this need not be the shortest-path metric of the induced subgraph on its fixed vertices.

Let $\Lambda \subseteq \mathbb{R}_{\geq 0}$ be the additive monoid generated by the positive distances in X , and let $D \subseteq \mathbb{R}$ be the additive group they generate. For a coefficient ring K , write $K[[q^\Lambda]]$ for formal sums $\sum_{\ell \in \Lambda} a_\ell q^\ell$, with multiplication $q^a q^b = q^{a+b}$. There are only finitely many elements of Λ in any bounded interval, so convolution is well-defined. If the metric is integral, this is naturally a subring of $K[[q]]$. All series below are formal; no analytic substitution for q is required. When comparing several spaces we use the monoid generated by all their distances.

For a commutative unital ring R , use the notation

$$C_{*,\ell}(X; R) = MC_{*,\ell}(X; R), \quad C^{*,\ell}(X; R) = \text{Hom}_R(C_{*,\ell}(X; R), R).$$

We assume the ordinary definitions and products, as in [HW17, LS21, He18, AI24], without recalling them. We need only name the standard basis set

$$\mathcal{T}_{n,\ell}(X) = \left\{ (x_0, \dots, x_n) : x_{i-1} \neq x_i, \quad \sum_{i=1}^n d(x_{i-1}, x_i) = \ell \right\}. \quad (1.1)$$

Thus $C_{n,\ell}(X; R) = R[\mathcal{T}_{n,\ell}(X)]$. If $\delta > 0$ is the least positive distance, then $\mathcal{T}_{n,\ell}(X) = \emptyset$ for $n > \ell/\delta$. The cases $|X| \leq 1$ satisfy the same boundedness assertion directly. Consequently every fixed-length chain complex is bounded and finite free over R . None of this requires a field.

We use $\text{Mag}(X; q)$ for ordinary formal magnitude. Its graded Euler characteristic description is the one in [HW17, Theorem 8] for graphs and [LS21, Corollary 7.15] for finite metric spaces. In particular, it is the alternating series of the cardinalities in (1.1), with integral coefficients. Over $\mathbb{Z}$, passing to ranks of homology has the same effect as tensoring homology with $\mathbb{Q}$; this numerical invariant does not record torsion.

1.2 Actions and the representation ring

The action on tuples is diagonal:

$$g(x_0, \dots, x_n) = (gx_0, \dots, gx_n). \quad (1.2)$$

It commutes with the magnitude differential, since isometries preserve the conditions determining that differential. On cochains the action is $(g\varphi)(c) = \varphi(g^{-1}c)$. Thus $\text{MH}_{n,\ell}(X; R)$ and $\text{MH}^{n,\ell}(X; R)$ are RG -modules, and the latter form an algebra on which G acts by bigraded automorphisms.

Source and status

This is the specialization of functoriality in [HW17, LS21, He18]; the automorphism-group action is explicitly noted in [HW17, §1.1]. The formulas also give a direct verification over every commutative R .

For the representation-valued Euler series we now take $R = \mathbb{C}$. The complex representation ring $R_{\mathbb{C}}(G)$ is the Grothendieck group of finite-dimensional complex representations, with $[U \oplus V] = [U] + [V]$ and multiplication $[U][V] = [U \otimes_{\mathbb{C}} V]$. Maschke's theorem identifies its additive group with the free abelian group on the irreducible representations. Its elements are *virtual representations*; their characters are defined by linear extension of $\chi_V(g) = \text{tr}(g | V)$. Character theory and its orthogonality formulas are used in their usual complex form [Se77, Chs. 1–2].

Definition 1.1. The representation-valued magnitude series is

$$\text{Mag}_G(X; q) = \sum_{\ell \in \Lambda} \sum_{n \geq 0} (-1)^n [\text{MH}_{n, \ell}(X; \mathbb{C})] q^\ell \in R_{\mathbb{C}}(G)[[q^\Lambda]]. \quad (1.3)$$

Each coefficient is a finite sum. The same series results from using cohomology: one may apply Euler–Poincaré to the dual permutation cochain complex. This assertion concerns the Euler series; no choice of an identification between individual homology and cohomology spaces is needed.

Source and status

Definition 1.1 is the equivariant specialization used in this note. Its analogy with equivariant Ehrhart theory is with the permutation representation on lattice points, whose character at g counts the lattice points fixed by g , as in [St11, §5, Lemma 5.2]. We do not attribute the present magnitude construction to that reference.

1.3 The character formula

Theorem 1.2 (Fixed-point character formula). *For $g \in G$, put $X^g = \{x \in X : gx = x\}$. Then*

$$\chi_{\text{Mag}_G(X; q)}(g) = \text{Mag}(X^g; q). \quad (1.4)$$

Proof. Fix a length ℓ . Euler–Poincaré for a finite complex with a chain endomorphism gives

$$\sum_n (-1)^n \text{tr}(g | \text{MH}_{n, \ell}(X; \mathbb{C})) = \sum_n (-1)^n \text{tr}(g | C_{n, \ell}(X; \mathbb{C})).$$

The trace of a permutation matrix counts fixed basis elements. By (1.2), a tuple is fixed by g exactly when every entry lies in X^g . Therefore the right side equals $\sum_n (-1)^n |\mathcal{T}_{n, \ell}(X^g)|$, the coefficient of q^ℓ in ordinary magnitude. This last identification is the ordinary Euler characteristic theorem cited above. The cochain argument gives the same result, since g and g^{-1} fix the same tuples. $\square$

Source and status

This theorem is proved here from the standard permutation trace identity and the ordinary magnitude Euler characteristic theorem. It is not an asserted degree-by-degree identification of homology of fixed spaces with fixed vectors in homology.

Corollary 1.3. *The following statements hold over $\mathbb{C}$.*

- (i) *Applying dimension to $\text{Mag}_G(X; q)$ gives $\text{Mag}(X; q)$.*
- (ii) *For $V \in \text{Irr}(G)$, the virtual multiplicity series of V is*

$$\sum_{n, \ell} (-1)^n \dim_{\mathbb{C}} \text{Hom}_G(V, \text{MH}_{n, \ell}(X; \mathbb{C})) q^\ell = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} \text{Mag}(X^g; q). \quad (1.5)$$

(iii) In particular, the invariant Euler series is

$$\sum_{n,\ell} (-1)^n \dim_{\mathbb{C}} \mathrm{MH}_{n,\ell}(X; \mathbb{C})^G q^\ell = \frac{1}{|G|} \sum_{g \in G} \mathrm{Mag}(X^g; q). \quad (1.6)$$

Proof. Evaluate (1.4) at 1, or take the character inner product with χ_V , respectively. For (iii), take $V = \mathbf{1}$. $\square$

The same arguments imply restriction compatibility for every subgroup $H \leq G$. They also give $\mathrm{Mag}_G(X \times Y; q) = \mathrm{Mag}_G(X; q) \mathrm{Mag}_G(Y; q)$ for the diagonal action and the sum metric. Indeed, fixed points commute with that product and ordinary magnitude is multiplicative: the magnitude matrix of the product is the Kronecker product of the two magnitude matrices. Injectivity of complex characters then gives the identity. These are deductions from Theorem 1.2.

Example 1.4 (Two points interchanged). Let $X = \{a, b\}$ with $d(a, b) = 1$, and let $G = C_2$ interchange them. There are two proper alternating tuples in degree n , of length n , and their differentials are zero. Thus, over every commutative R ,

$$\mathrm{MH}_{n,\ell}(X; R) = \begin{cases} R[C_2], & \ell = n, \\ 0, & \ell \neq n. \end{cases}$$

Over $\mathbb{C}$, writing ε for the sign representation,

$$\mathrm{Mag}_{C_2}(X; q) = \frac{[\mathbf{1}] + [\varepsilon]}{1 + q}.$$

Its character values are $2/(1 + q)$ and 0. Its invariant Euler series is $1/(1 + q)$, whereas $X^{C_2} = \emptyset$ and X/C_2 is a singleton of magnitude 1. The invariant chain $a + b$ explains why invariant vectors must not be confused with fixed points.

Remark 1.5 (Integral and modular information). The integral theory is the bigraded $\mathbb{Z}G$ -module $\mathrm{MH}_{*,*}(X; \mathbb{Z})$, including its torsion, rather than just (1.3). Integral homology groups need not be free abelian, so one cannot silently regard them as objects in a ring of integral lattices. Over a field of characteristic dividing $|G|$, magnitude homology still carries a group action, but semisimplicity and the above complex character reconstruction are unavailable. The series (1.3) is specifically a complex representation-ring invariant of the integral construction.

2 Borel equivariant magnitude (co)homology

2.1 Definition over a commutative ring

Coefficients

Throughout this subsection R is any commutative unital ring; in particular $R = \mathbb{Z}, \mathbb{Q}, \mathbb{C}$, or $\mathbb{F}_p$ is permitted. Give R the trivial RG -module structure. Derived tensor products and derived Hom are taken over the indicated ring, not over the ordinary magnitude homology groups.

Definition 2.1. For each length ℓ , define

$$\mathrm{MH}_{n,\ell}^{\mathrm{Bor},G}(X; R) = H_n(R \underset{RG}{\overset{\mathbf{L}}{\otimes}} C_{*,\ell}(X; R)), \quad (2.1)$$

$$\mathrm{MH}_{\mathrm{Bor},G}^{n,\ell}(X; R) = H^n(\mathbf{R}\mathrm{Hom}_{RG}(R, C^{*,\ell}(X; R))). \quad (2.2)$$

Here $H_n(G; M) = \text{Tor}_n^{RG}(R, M)$ and $H^n(G; M) = \text{Ext}_{RG}^n(R, M)$ denote group homology and cohomology. For example, a free group-bar resolution $Q_\bullet \rightarrow R$ gives the models

$$\text{Tot}(Q_\bullet \otimes_{RG} C_{*,\ell}), \quad \text{Tot Hom}_{RG}(Q_\bullet, C^{*,\ell}),$$

using a right resolution in the first formula and a left resolution in the second. The total differential has the usual tensor or Hom signs. Group degree contributes to total (co)homological degree, and has length zero.

Source and status

These are applications of the standard derived coinvariant and invariant constructions for group actions; see [Br82, Chs. I, III, VII]. We use them here as definitions for magnitude complexes. Their topological interpretation as reduced Borel constructions is given in §4.6, and is consistent with the usual Borel construction in [Gu20, §2.1.4].

The magnitude cochain product and a diagonal on the group-bar resolution equip $\text{MH}_{\text{Bor},G}^{*,*}$ with an associative bigraded product. All signs use total cohomological degree, not the length degree. Its algebraic identification with the Yoneda product is addressed in Theorem 3.2.

Proposition 2.2 (Group (co)homology spectral sequences). *Over every commutative R , there are first-quadrant spectral sequences*

$$H_p(G; \text{MH}_{q,\ell}(X; R)) \implies \text{MH}_{p+q,\ell}^{\text{Bor},G}(X; R), \quad (2.3)$$

$$H^p(G; \text{MH}^{q,\ell}(X; R)) \implies \text{MH}_{\text{Bor},G}^{p+q,\ell}(X; R). \quad (2.4)$$

The cohomological sequence, summed over lengths, is multiplicative.

Proof. Filter the above double complexes by group degree. The group-bar terms are finite free RG -modules because G is finite, so taking the magnitude differential first gives the displayed E_2 -pages. There are finitely many terms in each total degree, and the magnitude direction is bounded at every length. The standard first-quadrant convergence argument applies. Bar diagonals respect the filtration, which gives multiplicativity. This is the usual hyperhomology construction of [Br82, Ch. VII]; over fields the corresponding Ext-algebra statement is also [Ne14, Corollary 7.7]. $\square$

2.2 Comparison with the representation-valued theory

Proposition 2.3 (When the group order is invertible). *If $|G| \in R^\times$, then*

$$\text{MH}_{n,\ell}^{\text{Bor},G}(X; R) \cong \text{MH}_{n,\ell}(X; R)_G, \quad (2.5)$$

$$\text{MH}_{\text{Bor},G}^{n,\ell}(X; R) \cong \text{MH}^{n,\ell}(X; R)^G. \quad (2.6)$$

The averaging map identifies coinvariants with invariants in this situation. Over $\mathbb{C}$, the Borel Euler series equals

$$\frac{1}{|G|} \sum_{g \in G} \text{Mag}(X^g; q). \quad (2.7)$$

Proof. The idempotent $|G|^{-1} \sum_g g$ makes invariants an exact functor, and the trivial RG -module is projective on both sides. Higher group (co)homology vanishes, so Proposition 2.2 collapses. The isomorphism on coinvariants sends $[m]$ to $|G|^{-1} \sum_g gm$. The Euler series then follows from Corollary 1.3. $\square$

In (2.5), the subscript G on $\mathrm{MH}_{n,\ell}(X; R)_G$ means *coinvariants*:

$$M_G = M / \langle gm - m : g \in G, m \in M \rangle.$$

This is distinct from the superscript M^G of *invariants*, the fixed submodule. When $|G| \in R^\times$, averaging identifies the two, as stated in the proposition; over $\mathbb{Z}$ or in characteristic dividing $|G|$, they need not agree.

Remark 2.4 (What specializes, and what does not). For the trivial symmetry group both constructions recover ordinary magnitude (co)homology. For a nontrivial finite group in characteristic zero, Borel cohomology recovers the *invariant part* of the representation-valued theory. It does not recover the full representation from untwisted coefficients alone. Also, (2.7) need not equal either $\mathrm{Mag}(X^G; q)$ or $\mathrm{Mag}(X/G; q)$; Example 1.4 gives both failures.

There is a precise recovery statement if one allows coefficients twisted by irreducible representations. For a finite-dimensional complex representation V , put

$$\mathrm{MH}_{\mathrm{Bor},G}^{n,\ell}(X; V) := H^n \mathbf{R}\mathrm{Hom}_{\mathbb{C}G}(V, C^{*,\ell}(X; \mathbb{C})).$$

Equivalently this uses group cochains with values in $V^* \otimes C^{*,\ell}$. Semisimplicity gives

$$\mathrm{MH}_{\mathrm{Bor},G}^{n,\ell}(X; V) = \mathrm{Hom}_G(V, \mathrm{MH}^{n,\ell}(X; \mathbb{C})), \quad \mathrm{MH}^{n,\ell}(X; \mathbb{C}) \cong \bigoplus_{V \in \mathrm{Irr}(G)} V \otimes \mathrm{MH}_{\mathrm{Bor},G}^{n,\ell}(X; V). \quad (2.8)$$

The second is the evaluation isomorphism. This deduction uses the complex representation theory of [Se77, Ch. 2]; it is not a modular reconstruction assertion.

2.3 Stabilizers and modular phenomena

Proposition 2.5 (Length zero). *Over any commutative R ,*

$$\mathrm{MH}_{\mathrm{Bor},G}^{n,0}(X; R) \cong \bigoplus_{[x] \in X/G} H^n(G_x; R), \quad (2.9)$$

$$\mathrm{MH}_{n,0}^{\mathrm{Bor},G}(X; R) \cong \bigoplus_{[x] \in X/G} H_n(G_x; R). \quad (2.10)$$

Proof. The length-zero magnitude complex is $R[X]$ in degree zero. Decompose it into the permutation modules $R[G/G_x]$. For finite groups induction and coinduction of the trivial module give this same permutation module. Apply the homological and cohomological forms of Shapiro's lemma [Br82, Ch. III]. $\square$

For a single point, Borel magnitude (co)homology is group (co)homology, all in length zero. In particular, for $G = C_p$,

$$H^n(C_p; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & n = 0, \\ \mathbb{Z}/p, & n > 0 \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Over $\mathbb{F}_p$, $H^n(C_p; \mathbb{F}_p)$ is one-dimensional in every nonnegative degree. Both statements follow from the cyclic-group resolution alternating $g - 1$ and $1 + g + \cdots + g^{p-1}$ [Br82, Ch. I, §6]. Thus, in modular characteristic, even one fixed point can produce unbounded cohomological degree at a fixed length. The finite alternating sum used in Definition 1.1 is then unavailable; no Borel magnitude Euler series is asserted in this generality. Integral torsion is likewise invisible to a rank-only Euler characteristic.

If the action on vertices is free, every proper tuple has trivial stabilizer. Its chain and cochain modules are sums of free permutation RG -modules. These are acyclic for group homology and cohomology (by Shapiro), so the Borel complexes can be computed by ordinary coinvariant chains and invariant cochains even without inverting $|G|$. This is a chain-level statement, not a claim that group cohomology of each magnitude homology group vanishes. These quotient complexes still need not be the magnitude complex of the quotient metric space, as Example 1.4 shows.

Related literature

Fu–Yau construct a Borel version of *path* homology of directed graphs [FY26]. That is a related construction for a different homology theory; none of the magnitude comparison results here is being attributed to it.

3 Distance algebras and skew group constructions

3.1 The algebra, the two modules, and grading conventions

Coefficients

Unless a field hypothesis is explicitly added, R is an arbitrary commutative unital ring. Let

$$A = \Sigma_X(R), \quad J = A_{>0}, \quad S = A/J \cong \prod_{x \in X} R.$$

The distance algebra and the Asao–Ivanov identifications are assumed. We use b_{xy} for its standard basis and $e_x = b_{xx}$. The action $g(b_{xy}) = b_{gx,gy}$ preserves multiplication and grading, hence acts on A , J , and S . Write $\epsilon : A \rightarrow S$ for the quotient map.

Definition 3.1 (Skew group algebra). The *skew group algebra* is the R -module

$$B = A \rtimes_R G := A \otimes_R RG$$

with multiplication determined by

$$(a \otimes g)(b \otimes h) = a g(b) \otimes gh.$$

Put $u_g = 1 \otimes g$, give each u_g length zero, and set

$$T = B_0 = S \rtimes G = B/(J \rtimes G). \quad (3.1)$$

This is finite free of rank $|X|^2|G|$ over R . The module S has the following *separate* left and right B -structures:

$$(a \otimes g) \cdot s = \epsilon(a)g(s), \quad s \cdot (a \otimes g) = g^{-1}(s\epsilon(a)). \quad (3.2)$$

They are the structures used on the appropriate sides of Ext and Tor . We do not assert that these two actions commute to give a B -bimodule. For Ext we always use left modules. The left module S is the permutation module, not the regular T -module.

For a graded module, $M[\ell]_d = M_{d-\ell}$, and

$$\text{Ext}_A^{n,\ell}(M, N) = \text{Ext}_{D\text{-GrMod}(A)}^n(M, N[\ell]).$$

Tor uses its internal homogeneous component of degree ℓ . The same convention applies to B . In these conventions

$$\text{Tor}_{n,\ell}^A(S, S) \cong \text{MH}_{n,\ell}(X; R), \quad E_R := \text{Ext}_A^{*,*}(S, S) \cong \text{MH}^{*,*}(X; R). \quad (3.3)$$

The last is an algebra isomorphism with the magnitude cup product.

Source and status

These are [AI24, Theorems 5.3 and 5.6]; their Remark 5.7 explains the graded-opposite convention that would occur with right-module Ext . The equivariance in (3.3) follows directly from their bar models, whose tuple actions are (1.2). This is the only extra observation needed here.

3.2 Borel theory as ordinary Tor and Ext over the skew algebra

Theorem 3.2. *For every commutative R there are natural identifications*

$$\text{MH}_{\text{Bor},G}^{n,\ell}(X; R) \cong \text{Ext}_B^{n,\ell}(S, S), \quad (3.4)$$

$$\text{MH}_{n,\ell}^{\text{Bor},G}(X; R) \cong \text{Tor}_{n,\ell}^B(S, S). \quad (3.5)$$

The cohomological identification respects products. Equivalently, lengthwise there are derived comparisons

$$\mathbf{R}\text{Hom}_B(S, S) \simeq \mathbf{R}\text{Hom}_{RG}(R, \mathbf{R}\text{Hom}_A(S, S)), \quad (3.6)$$

$$S \underset{B}{\overset{\mathbf{L}}{\otimes}} S \simeq R \underset{RG}{\overset{\mathbf{L}}{\otimes}} (S \underset{A}{\overset{\mathbf{L}}{\otimes}} S). \quad (3.7)$$

The inner derived objects carry their natural G -actions.

Proof. Choose the equivariant left A -bar resolution $P_\bullet \rightarrow S$ from the left-module version of [AI24, Remark 5.4, §5.3]. It is A -projective and R -split as an augmented complex. Let $Q_\bullet \rightarrow R$ be a free left group-bar resolution. The total complex of $P_\bullet \otimes_R Q_\bullet$, with diagonal G -action, is a B -projective resolution of S . To see projectivity, for any equivariant A -module P there is an isomorphism

$$B \otimes_A P \longrightarrow P \otimes_R RG, \quad (au_g) \otimes p \longmapsto a g(p) \otimes g.$$

If P is A -projective, this is B -projective; each Q_p is free over RG . Exactness of the total augmentation follows from the R -split bar augmentations.

Tensor–Hom adjunction identifies the Hom complex of this resolution with

$$\text{Tot Hom}_{RG}(Q_\bullet, \text{Hom}_A(P_\bullet, S)).$$

Using (3.3) at the level of bar cochains gives (3.6). Bar diagonals and the usual lifting description of Yoneda composition identify its product with the Borel cup product. This is the bar-model mechanism of [Ne14, Theorem 7.5 and Corollary 7.7], where the general skew-group change-of-rings theorem is formulated over a field. The argument here works over R because the specified bars are R -split and all the induced terms are projective; no averaging or division is used.

For tensor products the degree-zero identity for equivariant right and left modules is $M \otimes_B N \cong (M \otimes_A N)_G$, where

$$g \cdot (m \otimes n) = m u_{g^{-1}} \otimes u_g n.$$

Resolving by the same induced projective terms gives (3.7). One can also verify acyclicity directly: for a free right B -module P , the G -module $P \otimes_A S$ is a sum of induced modules from the trivial subgroup, since S is finite free over R . It is therefore acyclic for coinvariants, and the standard derived-composition argument applies. The action above is the diagonal tuple action after (3.3). $\square$

This theorem realizes the two spectral sequences of Proposition 2.2 as skew-group change-of-rings spectral sequences. It also identifies the length-zero groups in Proposition 2.5 inside $\text{Ext}_B(S, S)$ and $\text{Tor}^B(S, S)$.

3.3 Recovery of the full equivariant theory

Proposition 3.3 (Induced coefficient module). *For every commutative R ,*

$$\mathrm{Ext}_B^{n,\ell}(T, S) \cong \mathrm{Ext}_A^{n,\ell}(S, S), \quad (3.8)$$

$$\mathrm{Tor}_{n,\ell}^B(T, S) \cong \mathrm{Tor}_{n,\ell}^A(S, S). \quad (3.9)$$

Both recover the full RG -module structure with the residual actions specified below.

Proof. There are canonical induction identifications

$${}_B T \cong B \otimes_A S, \quad T_B \cong S \otimes_A B.$$

Since B is free over A on both sides, derived induction–restriction adjunction proves the displayed formulas. These are the usual Eckmann–Shapiro comparisons, here proved simply by inducing a projective resolution.

For Ext , right multiplication $r_g : t \mapsto tu_g$ on the left module T induces, by contravariance, a left G -action on $\mathrm{Ext}_B(T, S)$. On $B \otimes_A P$ it lifts to

$$b \otimes p \longmapsto bu_g \otimes g^{-1}p.$$

Under adjunction its pullback sends a cochain f to $p \mapsto gf(g^{-1}p)$, precisely the action in ordinary equivariant cohomology. For Tor , left multiplication by u_g on the right module T lifts to $p \otimes b \mapsto gp \otimes u_gb$ on an induced right resolution. After tensoring with S , this acts diagonally on both factors and gives the ordinary equivariant homology action. $\square$

Proposition 3.4 (Twisted coefficients). *Let $R = \mathbb{C}$ and let V be a finite-dimensional representation of G . Give $S \otimes_{\mathbb{C}} V$ the diagonal action, with A acting on S . Then*

$$\mathrm{Ext}_B^{n,\ell}(S \otimes_{\mathbb{C}} V, S) \cong \mathrm{Hom}_G(V, \mathrm{MH}^{n,\ell}(X; \mathbb{C})).$$

In particular, these Ext groups recover all multiplicity spaces of the usual equivariant cohomology.

Proof. The derived comparison of Theorem 3.2, with first argument $S \otimes V$, identifies the result with group cohomology of $\mathrm{Hom}_{\mathbb{C}}(V, \mathbf{R}\mathrm{Hom}_A(S, S))$. Exactness of complex invariants gives the formula, which is (2.8) in algebraic form. $\square$

3.4 The Yoneda algebra and the role of simple modules

Theorem 3.5 (Yoneda skew-product identity). *For the distance-algebra situation above, over every commutative R ,*

$$\mathrm{Ext}_B^{*,*}(T, T) \cong E_R \rtimes G \cong \mathrm{MH}^{*,*}(X; R) \rtimes G \quad (3.10)$$

as bigraded algebras. Here T is the module in (3.1), whether or not it is semisimple.

Proof. Use an equivariant projective left A -bar resolution $P \rightarrow S$. Then $B \otimes_A P \rightarrow T$ is a projective left B -resolution. Equip $\mathrm{End}_A(P)$ with the conjugation G -action. There is an isomorphism of differential graded algebras

$$\mathrm{End}_A(P) \rtimes_R G = \mathrm{End}_A(P) \otimes_R RG \longrightarrow \mathrm{End}_B(B \otimes_A P), \quad \Phi(f \otimes g)(b \otimes p) = bu_g^{-1} \otimes f(gp). \quad (3.11)$$

Balancing follows from $au_g^{-1} = u_g^{-1}g(a)$ and A -linearity of f . Decomposing B into its finitely many group components shows that every endomorphism on the right has a unique expression of this form. Multiplication is checked directly:

$$\Phi(f \otimes g)\Phi(h \otimes t) = \Phi(f(g \cdot h) \otimes gt).$$

The group maps have degree zero and commute with the differentials, so (3.11) respects differential graded structures and internal lengths. Taking cohomology gives (3.10): the underlying complex of a skew product is a finite direct sum of copies of the original complex. Projective resolutions identify these endomorphism cohomologies with Yoneda Ext. No semisimplicity or division by $|G|$ occurs in this proof. $\square$

Source and status

Over a field with $\text{char } k \nmid |G|$, the standard general result is [MV01, Theorem 10]. The proof above establishes the version actually used here over arbitrary R , using the canonical equivariant bar resolution of S . It should not be read as a claim that T is the semisimple quotient in modular characteristic.

For example, the distinguished group elements retain the action by

$$(1 \otimes g)(\alpha \otimes 1)(1 \otimes g^{-1}) = (g\alpha) \otimes 1.$$

Thus recovery from (3.10) uses its natural group grading and distinguished copy of RG , rather than just an unmarked abstract algebra or its Morita class.

To discuss simples, take an algebraically closed field k . The ideal $J \rtimes G$ is nilpotent, so every simple B -module comes from T . Choosing a vertex representative in each orbit and transporters to the other vertices gives the groupoid-algebra decomposition

$$S \rtimes G \cong \prod_{[x] \in X/G} \text{Mat}_{|G_x|}(kG_x). \quad (3.12)$$

Indeed, a basis element $e_y u_g$ has source $g^{-1}y$ and target y ; its multiplication is composition of such arrows. Transporters identify arrows within an orbit with matrix units labelled by G_x , proving (3.12). Hence simple modules are indexed by pairs $([x], V)$ with V an irreducible kG_x -module. The permutation module S selects the trivial representation of each stabilizer.

If $\text{char } k \nmid |G|$, Maschke's theorem makes T semisimple and $T = B/\text{rad } B$. It contains every simple B -module with its regular-module multiplicity. In this case (3.10) is the usual Yoneda algebra of the semisimple quotient. Passing to one copy of each simple gives a full idempotent corner of this Ext algebra, not necessarily the identical algebra. In modular characteristic T can fail to be semisimple, although Theorems 3.2 and 3.5 remain valid with the specified modules.

3.5 Fixed-point spaces as corners

For a subgroup $H \leq G$, put $e_H = \sum_{x \in X^H} e_x$ and $S_H = e_H S$.

Proposition 3.6. *For every commutative R ,*

$$\Sigma_{X^H}(R) \cong e_H A e_H, \quad (e_H A e_H)_0 \cong S_H.$$

For $R = \mathbb{C}$ and $H = \langle g \rangle$, this gives the identity

$$\sum_{n, \ell} (-1)^n \text{tr}(g \mid \text{Ext}_A^{n, \ell}(S, S)) q^\ell = \sum_{n, \ell} (-1)^n \dim_{\mathbb{C}} \text{Ext}_{e_H A e_H}^{n, \ell}(S_H, S_H) q^\ell. \quad (3.13)$$

Proof. The corner has basis b_{xy} with $x, y \in X^H$, and its multiplication is exactly that for the restricted metric. The Ext identity follows from Theorem 1.2 and (3.3) applied to this metric space. If there are no fixed vertices, both the corner contribution and its magnitude groups are taken to be zero. $\square$

Source and status

The corner description is a direct deduction from the distance-algebra multiplication of [AI24, §5.1]; (3.13) is proved here by the character formula. Neither $e_H A e_H = A^H$ nor a general commutation of Ext with corners is asserted. The right side computes new Ext groups over the corner algebra, and (3.13) is an Euler identity.

Algebraic object	Interpretation and coefficients
$\text{Ext}_B(T, S)$ with residual action	Full usual equivariant cohomology; every commutative R .
$\text{Ext}_B(S, S)$	Borel cohomology; every commutative R . It is E_R^G when $ G \in R^\times$.
$\text{Ext}_B(T, T)$	$E_R \rtimes G$; every commutative R with $T = B_0$. It is the semisimple-quotient Yoneda algebra over non-modular fields.
$\text{Ext}_{e_H A e_H}(S_H, S_H)$	Cohomology of the restricted fixed-point metric space; every commutative R .

4 An integral Burnside-ring analogue

4.1 Finite group sets, orbits, and stabilizers

This section introduces the necessary Burnside-ring terminology from the beginning. The constructions in §§4.2–4.5 are integral: they use finite sets and integers and have no ground-field choice. Standard references for the background are [Bo00, §§2–3]; the metric-space applications are proved here.

A *finite G -set* is a finite set U with a left action of G . A map $f : U \rightarrow V$ is *equivariant* if $f(gu) = gf(u)$. An isomorphism of G -sets is an equivariant bijection. For $u \in U$, its *orbit* is $G_u = \{gu : g \in G\}$, and its *stabilizer* is $G_u = \{g : gu = u\}$. For a subgroup $H \leq G$,

$$U^H = \{u : hu = u \text{ for every } h \in H\}$$

is its set of *fixed points*. A set is *transitive* if it consists of one orbit. Every transitive G -set is isomorphic to G/H with the left coset action, for a stabilizer H , via $gH \mapsto gu$. Two such sets G/H and G/K are isomorphic exactly when H and K are conjugate. Consequently any finite G -set decomposes uniquely, up to order and isomorphism, into transitive orbits.

Write $\mathcal{S}(G)$ for a chosen set of representatives of the conjugacy classes of subgroups. A subgroup $N \leq G$ is *normal*, written $N \triangleleft G$, if $gNg^{-1} = N$ for every $g \in G$. Write $N_G(H) = \{g : gHg^{-1} = H\}$ for the *normalizer*, and $W_G(H) = N_G(H)/H$ for the *Weyl group*. These definitions apply to all subgroups, including 1 and G .

4.2 The Burnside ring and its structure maps

Definition 4.1 (Burnside ring). The *Burnside ring* $\mathcal{A}(G)$ is the abelian group generated by symbols $[U]$ for isomorphism classes of finite G -sets, subject to

$$[U \sqcup V] = [U] + [V].$$

Multiplication is $[U][V] = [U \times V]$, using the diagonal action. The zero is $[\emptyset]$ and the unit is $[G/G]$, a single fixed point. Arbitrary elements, including differences of sets, are called *virtual finite G -sets*.

Distributivity and commutativity follow from the corresponding equivariant bijections of finite sets. Orbit decomposition gives

$$\mathcal{A}(G) = \bigoplus_{H \in \mathcal{S}(G)} \mathbb{Z}[G/H]. \quad (4.1)$$

In particular, this is a free abelian group of finite rank. An element is represented by an actual finite set precisely when all its orbit coefficients are nonnegative. Such elements are sometimes called *effective*.

The product in this basis is described by double cosets:

$$[G/H][G/K] = \sum_{g \in H \backslash G / K} [G/(H \cap gKg^{-1})]. \quad (4.2)$$

Here $H \backslash G / K$ means the set of subsets HgK , and one representative g is chosen from each. To prove the formula, move the first coordinate of a pair of cosets to H . Its remaining orbits are represented by (H, gK) , whose stabilizer is $H \cap gKg^{-1}$. This is the standard orbit-product calculation of [Bo00, §§2.2–3.1].

The following maps will be useful.

- The *cardinality augmentation* $\text{aug} : \mathcal{A}(G) \rightarrow \mathbb{Z}$ sends $[U]$ to $|U|$. It is a ring homomorphism.
- *Restriction* $\text{Res}_H^G : \mathcal{A}(G) \rightarrow \mathcal{A}(H)$ forgets part of the action and is a ring homomorphism.
- *Induction* $\text{Ind}_H^G : \mathcal{A}(H) \rightarrow \mathcal{A}(G)$ sends $[V]$ to $[G \times_H V]$, where $(gh, v) \sim (g, hv)$. It is additive but generally not multiplicative. For $K \leq H$, it sends $[H/K]$ to $[G/K]$.
- *Conjugation* by g transports an H -action to a gHg^{-1} -action: ghg^{-1} acts as h did. It gives an isomorphism $c_g : \mathcal{A}(H) \rightarrow \mathcal{A}(gHg^{-1})$.

These maps satisfy transitivity, conjugation compatibility, the double-coset formula, and Frobenius reciprocity. For clarity, the last two identities are

$$\text{Res}_K^G \text{Ind}_H^G = \sum_{g \in K \backslash G / H} \text{Ind}_{K \cap gHg^{-1}}^K c_g \text{Res}_{g^{-1}Kg \cap H}^H, \quad (4.3)$$

$$a \text{Ind}_H^G(b) = \text{Ind}_H^G(\text{Res}_H^G(a)b). \quad (4.4)$$

They follow by decomposing the induced sets into K -orbits and by the bijection $(u, [g, v]) \mapsto [g, (g^{-1}u, v)]$, respectively. The compatible additive restriction, induction and conjugation data are called a *Mackey functor*; in the presence of ring structures and (4.4) they give the Burnside *Green functor*. We use these terms only for this concrete system of maps. The general definitions and these identities are in [Bo00, §2.2, Proposition 3.1.3, §5].

4.3 Marks and the table of marks

Definition 4.2 (Marks). For $H \leq G$, the H -*mark* is the ring homomorphism

$$m_H : \mathcal{A}(G) \longrightarrow \mathbb{Z}, \quad m_H([U]) = |U^H|.$$

It depends only on the conjugacy class of H . The combined *mark homomorphism* is

$$m : \mathcal{A}(G) \longrightarrow \prod_{H \in \mathcal{S}(G)} \mathbb{Z}, \quad a \longmapsto (m_H(a))_H. \quad (4.5)$$

The target is also called the *ghost ring*, with coordinatewise addition and multiplication.

The homomorphism property follows because fixed points commute with disjoint union and Cartesian product. Note that m_1 is cardinality, whereas m_G counts points fixed by the whole group.

Proposition 4.3 (Injectivity and rational reconstruction). *The map m is injective and has finite cokernel. After tensoring with $\mathbb{Q}$, it is an isomorphism*

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathcal{A}(G) \cong \prod_{H \in \mathcal{S}(G)} \mathbb{Q}.$$

Proof. In the orbit basis the matrix is the [table of marks](#)

$$M_{H,K} = m_H([G/K]) = |\{gK : g^{-1}Hg \leq K\}|. \quad (4.6)$$

An entry can be nonzero only if H is conjugate to a subgroup of K . Order the subgroup representatives by nonincreasing order. The matrix is triangular, with diagonal entry $M_{H,H} = |N_G(H) : H| = |W_G(H)|$, a positive integer. It has nonzero determinant, which proves all assertions. $\square$

Source and status

This is the classical mark theorem, with the standard triangular-matrix proof; see [Bo00, §3.2]. In particular, arbitrary integer lists of fixed-point counts need not come from a virtual finite set: they must belong to the integral image of this matrix.

Example 4.4 (A cyclic group of prime order). For $G = C_p$, put $1 = [G/G]$ and $t = [G/1]$. Then

$$\mathcal{A}(C_p) = \mathbb{Z}1 \oplus \mathbb{Z}t, \quad t^2 = pt,$$

and

$$m_G(a + bt) = a, \quad m_1(a + bt) = a + pb.$$

Thus an integer pair (z_1, z_G) is a mark vector precisely when $z_1 \equiv z_G \pmod{p}$. This is a direct calculation from the two orbit types, not an additional assumption on marks.

4.4 Linearization and information lost in characters

Definition 4.5 (Complex linearization). The ring homomorphism

$$L_{\mathbb{C}} : \mathcal{A}(G) \longrightarrow R_{\mathbb{C}}(G), \quad [U] \longmapsto [\mathbb{C}[U]]$$

is called [linearization](#). Its character satisfies

$$\chi_{L_{\mathbb{C}}(a)}(g) = m_{\langle g \rangle}(a). \quad (4.7)$$

Indeed, the trace counts elements of U fixed by g . Thus complex characters see exactly the marks at cyclic subgroups. In particular, $a \in \ker L_{\mathbb{C}}$ if and only if all its cyclic-subgroup marks vanish. This is the usual permutation-character construction [Se77, Ch. 1], combined with the definition of marks. It does not imply that every complex representation is a linearized finite set.

Example 4.6 (A nonzero element killed by linearization). Let $G = C_2 \times C_2$, with its three order-two subgroups H_1, H_2, H_3 . The element

$$a = [G/1] - \sum_{i=1}^3 [G/H_i] + 2[G/G]$$

has $m_1(a) = 4 - 6 + 2 = 0$ and $m_{H_i}(a) = 0 - 2 + 2 = 0$ for every i . These are all cyclic subgroups, so $L_{\mathbb{C}}(a) = 0$. However, $m_G(a) = 2$, so $a \neq 0$. This calculation shows explicitly how whole-subgroup fixed-point information can be lost upon passing to complex representations.

For every commutative R the operation $U \mapsto R[U]$ is still available on finite sets. The comparison with ordinary complex characters in (4.7) specifically uses $R = \mathbb{C}$. We make no identification of $\mathcal{A}(G)$ with a ring of RG -modules: linearization can lose information even over $\mathbb{C}$.

4.5 The Burnside-valued magnitude series

Definition 4.7. The integral Burnside-valued magnitude series is

$$\text{Mag}_G^A(X; q) = \sum_{\ell \in \Lambda} \sum_{n \geq 0} (-1)^n [\mathcal{T}_{n, \ell}(X)] q^\ell \in \mathcal{A}(G)[[q^\Lambda]]. \quad (4.8)$$

The brackets here mean finite G -sets, not modules or homology groups. The boundedness established in §1 makes every coefficient well-defined integrally.

Theorem 4.8 (Marks and linearization of magnitude). *For every subgroup $H \leq G$,*

$$m_H(\text{Mag}_G^A(X; q)) = \text{Mag}(X^H; q). \quad (4.9)$$

Moreover,

$$L_{\mathbb{C}}(\text{Mag}_G^A(X; q)) = \text{Mag}_G(X; q). \quad (4.10)$$

Thus the collection of ordinary magnitudes of all restricted fixed-point spaces uniquely determines the Burnside-valued series.

Proof. Coordinatewise fixedness gives $\mathcal{T}_{n, \ell}(X)^H = \mathcal{T}_{n, \ell}(X^H)$. Applying a mark to (4.8) therefore gives the ordinary chain Euler series, proving (4.9). Applying linearization gives the virtual Euler characteristic of the complex permutation chain modules. Euler–Poincaré in $R_{\mathbb{C}}(G)$ gives (4.10). Uniqueness follows coefficientwise from Proposition 4.3. $\square$

Source and status

This theorem is proved here by applying the standard Burnside-ring constructions to the magnitude tuple sets. The only magnitude input is the ordinary Euler characteristic theorem [HW17, LS21]. In particular, existence is established integrally by (4.8); one does not merely postulate a rational mark vector and assume it has an integral lift.

Several consequences follow immediately.

- (i) Cardinality recovers $\text{Mag}(X; q)$, while the whole-group mark recovers $\text{Mag}(X^G; q)$.
- (ii) Restriction is compatible with restricting the group action. The sum-metric product satisfies

$$\text{Mag}_G^A(X \times Y; q) = \text{Mag}_G^A(X; q) \text{Mag}_G^A(Y; q).$$

For the product assertion apply every mark, use multiplicativity of ordinary magnitude, and use injectivity of marks.

- (iii) For $G = C_p$, Example 4.4 gives the coefficientwise integral congruence

$$\text{Mag}(X; q) \equiv \text{Mag}(X^{C_p}; q) \pmod{p}. \quad (4.11)$$

Indeed,

$$\text{Mag}_{C_p}^A(X; q) = \text{Mag}(X^{C_p}; q) 1 + \frac{\text{Mag}(X; q) - \text{Mag}(X^{C_p}; q)}{p} t,$$

and the quotient has integral coefficients.

(iv) If G acts freely on X , every tuple set is free, so

$$\text{Mag}_G^A(X; q) = [G/1] \frac{\text{Mag}(X; q)}{|G|},$$

where the quotient is again coefficientwise integral. This does not identify it with magnitude of the quotient metric space.

All four are deductions from the construction and mark theorem. The noninjectivity of linearization shows that the Burnside target can retain more information than the representation-ring target; it does not by itself supply two metric examples distinguished by their Burnside magnitude but not their representation-valued magnitude.

4.6 An equivariant topological model

To explain a possible homological refinement, we use the standard pointed simplicial magnitude model, solely to expose its fixed-point structure. For graphs this is [HW17, Definition 40 and Lemma 42]; the same description for finite metric spaces follows verbatim from the length condition and is compatible with [LS21].

For a length ℓ , let $N_\ell(X)$ have a basepoint in each simplicial degree and, in degree n , the tuples of total length ℓ , now allowing consecutive repetitions. A face deletes an entry if length is preserved and otherwise gives the basepoint; a degeneracy repeats an entry. Put $K_\ell(X) = |N_\ell(X)|$, its pointed geometric realization. Its non-basepoint nondegenerate n -simplices are precisely $\mathcal{T}_{n,\ell}(X)$, and its reduced normalized chain complex is $C_{*,\ell}(X; \mathbb{Z})$.

A *G-CW complex* is built by attaching cells $G/H \times D^n$, with G acting on the coset factor and trivially on the disk, along their boundaries. In this model, the stabilizer of a tuple acts trivially on its simplex: it fixes every vertex position. Thus $K_\ell(X)$ is a finite pointed G -CW complex. It is finite because only finitely many nondegenerate tuples have the given length.

Proposition 4.9. *For every $H \leq G$,*

$$N_\ell(X)^H = N_\ell(X^H), \quad K_\ell(X)^H \cong K_\ell(X^H).$$

Consequently, for every commutative R , $\tilde{H}_n(K_\ell(X)^H; R) \cong \text{MH}_{n,\ell}(X^H; R)$.

Proof. The simplicial identity is coordinatewise fixedness, including degeneracies and the basepoint. In a geometric realization every point is represented in the interior of a unique nondegenerate simplex. If it is fixed by H , this simplex is stabilized by H ; in the present model the stabilizer fixes it pointwise. This proves the realization identity. Reduced cellular chains give the final claim. $\square$

For a finite pointed G -CW complex K , let $\mathcal{E}_n$ be its finite G -set of non-basepoint n -cells. Its *reduced equivariant Euler characteristic* is

$$\tilde{\chi}^G(K) = \sum_n (-1)^n [\mathcal{E}_n] \in \mathcal{A}(G).$$

Every mark is $\tilde{\chi}(K^H)$, so injectivity of marks shows that this expression is independent of the chosen finite G -CW structure and invariant under equivariant homotopy equivalence. This is the usual equivariant Euler construction, in the form of [Bo00, §4], with the basepoint subtracted. Here it gives

$$\text{Mag}_G^A(X; q) = \sum_\ell \tilde{\chi}^G(K_\ell(X)) q^\ell. \quad (4.12)$$

For comparison with §2, choose a contractible free G -CW complex EG and put $BG = EG/G$. The usual reduced Borel model is the pair

$$(EG \times_G K_\ell(X), EG \times_G \{*\}).$$

Its relative chains are modeled by $C_*(EG; R) \otimes_{RG} \tilde{C}_*(K_\ell(X); R)$, using cellular or simplicial Eilenberg–Zilber. They compute (2.1); relative cochains compute (2.2). The basepoint subspace is BG and must be removed in this relative/reduced construction. For X a point, $K_0(X) = S^0$, so this still yields the ordinary (unreduced) group (co)homology of BG , as in §2. This explains the Borel terminology and the reduction convention; see [Gu20, §2.1.4] for the underlying topological construction.

4.7 A possible Bredon refinement: definitions and scope

The pointed complexes above allow one to apply Bredon (co)homology. We describe the necessary terminology explicitly. The standard construction is in [Gu20, §2.1.6]; its application to $K_\ell(X)$ is the proposal made here.

Definition 4.10 (Orbit category and coefficient systems). The *orbit category* $\mathcal{O}_G$ has objects G/H , one for each subgroup, and morphisms the equivariant maps between these sets. A *contravariant coefficient system over R* is a functor $\mathcal{O}_G^{\text{op}} \rightarrow R\text{-Mod}$. A *covariant coefficient system* is a functor $\mathcal{O}_G \rightarrow R\text{-Mod}$.

A map $f_g : G/H \rightarrow G/K$ is determined by $f_g(H) = gK$, and exists precisely when $g^{-1}Hg \leq K$. It induces a map $K_\ell(X)^K \rightarrow K_\ell(X)^H$, $z \mapsto gz$. Thus

$$\underline{C}_{n,\ell}(X; R)(G/H) = \tilde{C}_n(K_\ell(X)^H; R) = R[\mathcal{T}_{n,\ell}(X^H)] \quad (4.13)$$

is a chain complex of contravariant coefficient systems. The chains here are generated by fixed cells, not invariant linear combinations of all cells. A cell orbit G/K contributes the representable system $R[\text{Hom}_{\mathcal{O}_G}(-, G/K)]$. Hence this is precisely the usual reduced Bredon cellular chain system. Its objectwise homology is the diagram $G/H \mapsto \text{MH}_{n,\ell}(X^H; R)$.

For a covariant system M , define the tensor product over the orbit category by

$$\underline{C}_{n,\ell} \otimes_{\mathcal{O}_G} M = \left(\bigoplus_H \underline{C}_{n,\ell}(G/H) \otimes_R M(G/H) \right) / \langle \underline{C}(f)c \otimes m - c \otimes M(f)m \rangle. \quad (4.14)$$

Here $f : G/H \rightarrow G/K$, $c \in \underline{C}_{n,\ell}(G/K)$, and $m \in M(G/H)$. This quotient is often called a *coend*; the displayed relations are its full meaning here. For a contravariant system N , let $\text{Nat}_R(\underline{C}_{n,\ell}, N)$ denote the R -module of families of R -linear maps commuting with all orbit-category morphisms.

Definition 4.11 (Bredon magnitude groups). For any commutative R and coefficient systems as above, put

$$\text{MH}_{n,\ell}^{\text{Br},G}(X; M) = H_n(\underline{C}_{*,\ell}(X; R) \otimes_{\mathcal{O}_G} M), \quad (4.15)$$

$$\text{MH}_{\text{Br},G}^{n,\ell}(X; N) = H^n(\text{Nat}_R(\underline{C}_{*,\ell}(X; R), N)). \quad (4.16)$$

These are ordinary reduced Bredon groups of $K_\ell(X)$, with length retained. When $G = 1$ and the coefficient system is R , they reduce to ordinary magnitude (co)homology. Their equivariant homotopy invariance as invariants of $K_\ell(X)$ is the usual Bredon invariance, not a new assertion of metric homotopy invariance.

One natural integral choice uses the Burnside coefficient systems: at G/H take $\mathcal{A}(H)$. On $f_g : G/H \rightarrow G/K$, the covariant structure map is

$$\mathcal{A}(H) \xrightarrow{c_{g^{-1}}} \mathcal{A}(g^{-1}Hg) \xrightarrow{\text{Ind}_{g^{-1}Hg}^K} \mathcal{A}(K),$$

and the contravariant structure map is

$$\mathcal{A}(K) \xrightarrow{\text{Res}_{g^{-1}Hg}^K} \mathcal{A}(g^{-1}Hg) \xrightarrow{c_g} \mathcal{A}(H).$$

These are well-defined on the coset gK and respect composition; inner conjugations act trivially on isomorphism classes of group sets. They are the covariant and contravariant parts of the Burnside Mackey functor introduced above. Over a general commutative R , replace them by $R \otimes_{\mathbb{Z}} \mathcal{A}(H)$.

What is established and what remains a refinement problem

The integral series (4.8), its mark formula, its linearization, and its realization as (4.12) are established by the proofs in this section. The Bredon groups are also well-defined by the displayed construction. However, a Bredon homology group is an abelian group (or part of a coefficient system), not automatically an element of $\mathcal{A}(G)$. We have not identified an alternating sum of their classes in some unspecified Grothendieck group with (4.8). Nor does the ordinary $\mathbb{Z}G$ -module $\text{MH}_{n,\ell}(X; \mathbb{Z})$ have a canonical Burnside class: homology need not possess a permutation basis, and linearization is not injective.

The orbit-category construction retains all subgroup fixed-point complexes and the maps between them. The skew group algebra in §3 organizes equivariant A -modules and the Borel/representation comparisons, while the individual fixed-point distance algebras $e_H A e_H$ realize the values at subgroups. A further algebraic description of the complete orbit-category construction, including its restriction and induction structures, is a reasonable next problem. No identification of that entire structure with ordinary Tor or Ext over the single skew group algebra is claimed here.

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